

A Multiplicity Counterexample to a Uniform Many-Polynomial Bézout Estimate

Automatic mathematics run `fa_banach_001`

9 August 2026

Status. Candidate counterexample, likely valid, for human review. It gives a full negative answer to the open problem in Section 10.4 of Fricain–Hartmann–Ross–Timotin as the problem is stated.

1 Source problem

Fricain, Hartmann, Ross, and Timotin study coefficient-norm estimates for polynomial Bézout identities [1]. For $A_1, \dots, A_q \in \mathbb{C}[z]$ with no common zero, they define

$$\delta = \min \left\{ \sum_{i=1}^q |A_i(z)| : z \in \mathbb{C}, \prod_{i=1}^q A_i(z) = 0 \right\}.$$

Their Section 10.4 asks for a constant $C > 0$, depending only on $\max_i \deg A_i$, such that one can solve

$$\sum_{i=1}^q A_i R_i = 1 \quad \text{with} \quad \max_i \|R_i\| \leq \frac{C}{\delta^2}, \tag{1}$$

where $\|R\|$ is the maximum modulus of the coefficients of R . The preceding paragraph also gives the usual degree restriction $\deg R_i \leq \max_j \deg A_j - 1$.

The complete source page follows. Notice especially that C may depend on the maximal degree but not on the number q of polynomials.

10.3. Varying the degrees of A, B . As stated from the beginning, the various constants that appear in our results depend only on $N = \deg A$ and $K = \deg B$ (and not the coefficients of A and B). If we let N and K increase, one can see that our methods produce constants that increase quite rapidly in N and K (see (7.6) for example). For possible numerical applications, it would be interesting to explore the optimal estimates of these constants as functions of N and K .

10.4. Generalizing to more than two polynomials. Bézout's polynomial identity from (1.2) generalizes to more than two polynomials in that if $A_1, \dots, A_q \in \mathbb{C}[z]$ have no common zeros, then there exist $R_1, \dots, R_q \in \mathbb{C}[z]$ with $\deg R_i \leq \max\{\deg A_j\} - 1$ such that $A_1 R_1 + \dots + A_q R_q = 1$ (see, for instance, [15] Section 1.3). However, our methods do not extend to provide estimates of $\|R_i\|$ when $q \geq 3$. Note that when $q \geq 3$, the solutions R_i are no longer unique, even if we put restrictions on their degree.

Open problem. Suppose that $A_1, \dots, A_q \in \mathbb{C}[z]$ have no common zeros, and define

$$\delta = \min \left\{ \sum_{i=1}^q |A_i(z)| : z \in \mathbb{C}, \prod_{i=1}^q A_i(z) = 0 \right\}.$$

Prove there exists a constant $C > 0$, depending only on $\max\{\deg A_i\}$, such that one can find $R_1, \dots, R_q \in \mathbb{C}[z]$ with $A_1 R_1 + \dots + A_q R_q = 1$ and

$$\max_{0 \leq i \leq q} \|R_i\| \leq \frac{C}{\delta^2}.$$

ACKNOWLEDGEMENTS

We are indebted to Laurent Baratchart for pointing out the problem of estimating the norm of the inverse of the Sylvester matrix.

Emmanuel Fricain was supported the Labex CEMPI (ANR-11-LABX - 0007-01). Andreas Hartmann was supported by the Project REPKA (ANR-18-CE40-0035). Dan Timotin was partially supported by a grant of the Ministry of Research, Innovation and Digitization, CNCS/CCCDI – UEFISCDI, Project Number PN-III-P4-ID-PCE-2020-0458, within PNCDI III, and by the international research network ECO-Math.

REFERENCES

- [1] Carlos A. Berenstein, Roger Gay, Alekos Vidras, and Alain Yger. *Residue currents and Bezout identities*, volume 114 of *Progress in Mathematics*. Birkhäuser Verlag, Basel, 1993.
- [2] Carlos A. Berenstein and Alain Yger. Effective Bezout identities in $\mathbb{Q}[z_1, \dots, z_n]$. *Acta Math.*, 166(1-2):69–120, 1991.
- [3] Étienne Bézout. *Théorie générale des équations algébrique*. Paris, France, 1779.

2 Counterexample

The obstruction is duplication. The definition of δ adds the moduli of all inputs, so repeating generators makes δ grow linearly. But repeating a generator can reduce the largest Bézout coefficient only linearly, whereas (1) demands a quadratic reduction.

Theorem 1. *There is no constant in (1) depending only on the maximal degree. This remains false under the normalization $\|A_i\| \leq 1$, and remains false even if the solution polynomials R_i are allowed arbitrary degrees.*

Proof. Let $m \geq 1$, put $q = 2m$, and define

$$A_1 = \cdots = A_m = z, \quad A_{m+1} = \cdots = A_{2m} = 1 - z. \quad (2)$$

These polynomials have no common zero, their maximal degree is one, and every one of them has coefficient norm exactly one.

The product in the definition of δ vanishes precisely at $z = 0$ and $z = 1$. At either point exactly m summands in $\sum_i |A_i(z)|$ equal one and the other m vanish. Consequently

$$\delta = m. \quad (3)$$

Consider any polynomial solution $R_1, \dots, R_{2m}$ of the Bézout identity for (2). Evaluating it at $z = 0$ gives

$$\sum_{i=m+1}^{2m} R_i(0) = 1. \quad (4)$$

It follows from the triangle inequality that

$$1 \leq \sum_{i=m+1}^{2m} |R_i(0)| \leq m \max_{m < i \leq 2m} \|R_i\|,$$

and hence

$$\max_{1 \leq i \leq 2m} \|R_i\| \geq \frac{1}{m}. \quad (5)$$

This argument uses only constant terms, so allowing higher degrees does not help. (Evaluation at $z = 1$ gives the same lower bound on the first block.)

If (1) held, maximal degree one would determine a single constant C , independent of m . Equations (3) and (5) would give

$$\frac{1}{m} \leq \max_i \|R_i\| \leq \frac{C}{m^2}.$$

Choosing an integer $m > C$ is a contradiction. □

3 Exact obstruction and necessary repair

The preceding lower bound is not an artifact of the proof.

Proposition 2. *For the family (2), the minimum possible value of $\max_i \|R_i\|$, over all polynomial Bézout solutions, is exactly*

$$\frac{1}{m} = \frac{q}{2\delta^2}. \quad (6)$$

Proof. The lower bound is (5). For equality, take the constant polynomial $R_i = 1/m$ for every i . Then

$$\sum_{i=1}^{2m} A_i R_i = mz \frac{1}{m} + m(1-z) \frac{1}{m} = 1.$$

Since $q = 2m$ and $\delta = m$, this optimum equals the right-hand side of (6). $\square$

Thus a repaired estimate of the form $C(q, d)\delta^{-2}$ must have $C(q, 1) \geq q/2$ along even values of q . Equivalently, if the ℓ^1 -type separation parameter in the source is retained, at least linear dependence on the number of generators is unavoidable. The example does not decide whether a bound with linear q -dependence is sufficient in general, nor does it address the sharp dependence on δ when q is fixed.

4 Verification and novelty scope

The proof reduces to the exact identities (3)–(5). An independent script checks the source parameter, the explicit extremal solution, and (6) with exact rational arithmetic for $1 \leq m \leq 10,000$. The computation is a sanity check and is not used in the proof.

A bounded literature audit was performed on 9 August 2026 after the direct argument: the run indexes, exact-phrase and core-keyword web searches, the current arXiv record, and author/citation search pages were checked. No later answer or occurrence of this duplication example was found. Novelty is therefore plausible, not certified.

References

- [1] E. Fricain, A. Hartmann, W. T. Ross, and D. Timotin, *An analytic approach to estimating the solutions of Bézout’s polynomial identity*, arXiv:2310.12734v3 (2024).